Operational Training Manual & Forensic Reference Guide

Design A: Legacy Statistical Workspace

Insider Threat Intelligence Hub • Interactive Evaluation Study

Monitored Target: User 162 (`AYG1697`) **Baseline Context:** 90 Days **Schedule:** Mon–Fri, 08:00–18:00
Neural Anomaly Cutoff: 0.0742 **Task Protocol:** Two-Alert Walkthrough **Mode:** Design A (Legacy)

Institutional Research Ethics Statement & Voluntary Participation Protocol

This training and evaluation system operates under university-approved Office of Research Ethics (ORE) guidelines. **Participation is completely voluntary.** You retain full autonomy to decline answering any question or skip specific evaluative stages at any time without penalty.
Because the SOC platform relies on an automated database submission pipeline to compile experimental records and unlock the evidence export function, participants exercising their option to pass may utilize standardized structural bypass entries:

- **Dropdown Selections:** Select **"Prefer not to answer"** or **"Pass"** from the disposition and classification menus.
- **Text Input Fields:** Enter **"Pass"** or **"N/A"** into the Analyst Case Justification Narrative or interface reflection boxes.
- **Workload Reflection (NASA-TLX):** Leave the Mental Demand and Frustration sliders at their default midpoint values (11).

1. Workspace Architecture & Interface Layout

Design A provides a traditional statistical operations environment. Upon system initialization, analysts access tools organized into structured functional modules:

- **Top Header Bar:** Houses the monitored user credentials (`User 162 / AYG1697`), the 90-day baseline indicator, and expected user schedule badges (Monday through Friday, 08:00 to 18:00).
- **Architecture Selector:** Located toward the left of the header; allows toggle verification between **Design A (Legacy View)** and **Design B (XAI Suite)**. Ensure Design A is active.
- **Model Benchmark Indicators:**
  - **Anomaly Cutoff Line (**0.0742**):** The empirical threshold determining alert triggers.
  - **Recall (True Catch Rate):** Proportion of historical ground-truth threat cases successfully flagged during validation.
  - **Precision Vector:** Proportion of flagged alerts that corresponded to verified threats during validation.
  - *Operational Principle:* Global model metrics provide overall system confidence context; they do not substantiate maliciousness for any single isolated alert.
- **KPI Audit Strip:** Summarizes ongoing triage throughput across four audit telemetry cards: *Baseline Context Size*, *Audited Window*, *Triggered Incidents*, and *Evaluated Audits*.
- **Date-wise Anomaly Score Timeline:** Plots daily autoencoder reconstruction error curves against the red anomaly cutoff line (0.0742). Any calendar coordinate breaching this line generates an active investigation row.

2. Statistical Framework: Macro vs. Micro Z-Scores

Design A distinguishes between aggregate daily error anomalies and individual behavioral feature shifts using dual Z-score indicators.

2.1 Macro Severity: Day-Level Reconstruction Error Z-Score

Accessed via the **"Show Day-Level Reconstruction Error Z-scores"** toggle in the upper-right corner of the *Detected Anomalies* table header. When toggled off, the table presents qualitative labels; when toggled on, it displays signed standard deviations relative to the baseline error distribution:

- **Stable Baseline Profile ($Z \leq 2.0$):** Expected statistical variation within standard error margins.
- **Significant Anomaly Drift ($2.0 < Z \leq 4.5$):** Moderate deviation from normal operating norms.
- **Severe Outlier Event ($Z > 4.5$):** Extreme divergence requiring immediate forensic review.

2.2 Micro Severity: Feature-Level Behavioral Deviation Z-Scores

Accessed via the **"Show Feature-Level Behavioral Deviation Z-scores"** toggle inside the *Autoencoder Input Features* panel. Calculates standardized deviance for specific behavioral actions:

$$\text{Feature } Z = \frac{|\text{Observed Value} - \mu_{\text{baseline}}|}{\sigma_{\text{baseline}}}$$

- **Within Baseline Range:** $Z < 2.0\sigma$ (Standard operational behavior; dismiss as safe).
- **Moderate Feature Deviation:** $2.0\sigma \leq Z \leq 3.0\sigma$ (Elevated feature activity).
- **Extreme Feature Deviation:** $Z > 3.0\sigma$ (Substantial behavioral outlier).

Critical Analytic Rule: Statistical Anomaly vs. Malicious Intent

A large positive Z-score confirms statistical rarity, not malicious tradecraft. Neither macro nor micro Z-scores establish adverse intent on their own. Threat validation requires matching the anomalous features to structured threat patterns and confirming an unbroken execution sequence in the raw event logs.

3. Threat Checklist & Scenario Matching Rules

Analyses must adhere to the formal **Hierarchical Triage Principle**:

1. **Screen the Primary Indicator:** Identify whether the mandatory primary signature is present. If the primary signature is absent, that scenario cannot be confirmed. Never treat isolated secondary flags as a positive match.
2. **Cross-Check Secondary Indicators:** If the primary signature is verified, examine designated secondary flags to corroborate malicious intent.
3. **Audit Raw Chronological Logs:** Validate timestamps, data targets, and user commands in the *Raw User Actions* table to verify a complete, coherent attack vector.

Code	Scenario Name	Primary Indicator (Mandatory)	Secondary Indicators (Corroborating)
S-1	IP Theft via Media	**Wikileaks Web Access**	Thumb Drive Usage, After-Hours Logon
S-2	Fraud via Job & USB	**Job Site Browsing**	Competitor Email Traffic, Thumb Drive Usage
S-3	Misuse via Keylogger	**Keylogger Software Download**	Unauthorized Host Login, Mass External Email
S-4	Data Exfiltration	**Unauthorized Host Login**	Deep File System Search, Personal Webmail
S-5	Mixed Profile	**Overlapping Multi-Vector Spikes**	Synchronous cross-vector volume anomalies
S-6	Benign Anomaly	**Isolated Non-Threat Deviation**	Workload spikes, safe shift changes; no exfiltration

4. Standard 6-Step Investigation Workflow

1. **Case Identification:** In the *Master Triage Control Panel*, select the target date row within the *Detected Anomalies* table to bind the contextual panels to that date.
2. **Macro Evaluation:** Review the anomaly score's relation to the 0.0742 threshold. Toggle day-level reconstruction error Z-scores to record global statistical variance.
3. **Checklist Screening:** Cross-reference flagged behaviors against Scenarios S-1 through S-6 using the mandatory primary indicator filter.
4. **Micro Audit:** Examine the *Autoencoder Input Features* panel. Activate the feature-level deviation toggle to isolate which specific behaviors register moderate (2σ–3σ) or extreme ($> 3\sigma$) departures.
5. **Chronological Log Review:** Open the *Raw User Actions* panel. Scrutinize precise timestamps, external URLs, file paths, and removable drive connect/disconnect sequences.
6. **Disposition Logging:** Assign an *Operational Disposition* (`Confirm Threat`, `Dismiss Alarm`, or `Prefer not to answer`) and assign the appropriate *Day Scenario* (S-1 through S-6) in the table row.

5. Guided Alert Walkthroughs

5.1 Training Alert 1: October 13, 2010 (Dismiss Alarm / Benign Anomaly)

- **Telemetry:** Anomaly magnitude breaches the 0.0742 cutoff; requires investigation.
- **Primary Indicator Screening:** Web activity reveals elevated job search portal access, activating **S-2 (Fraud via Job & USB)** as a tentative candidate. Primary indicators for S-1 (Wikileaks), S-3 (Keylogger), and S-4 (Unauthorized Login) are absent.
- **Secondary Indicator Audit:** Autoencoder input metrics reveal competitor email contacts; however, **Thumb Drive activity is zero**. File transfer volume and night-time activity metrics remain completely stable within normal baseline margins.
- **Raw Action Log Verification:** The user logs in and logs out strictly within normal working hours (08 : 00–18 : 00). There is no removable device mount, no local directory staging, and no data egress sequence.
- **Analytic Finding:** The event represents an isolated behavioral variance without a malicious exfiltration chain.
- **Logged Decision:**
  - **Operational Disposition:** `Dismiss Alarm`
  - **Day Scenario:** `S-6 (Benign Anomaly)`

5.2 Training Alert 2: October 26, 2010 (Confirm Threat / IP Theft via Media)

- **Telemetry:** Anomaly score significantly breaches the threshold (0.11602 vs. 0.0742 cutoff); classified under a *Severe Outlier* track.
- **Primary Indicator Screening:** Web monitoring confirms direct outbound HTTP queries to `wikileaks.org`, satisfying the mandatory primary indicator for **S-1 (IP Theft via Media)**.
- **Secondary Indicator Audit:** Feature Z-scores display extreme deviations ($> 3\sigma$) in **Thumb Drive Usage Count** and **After-Hours Activity Count**.
- **Raw Action Log Verification:** Shortly past midnight (outside standard operating hours), sequential access to sensitive repositories occurs. This is immediately followed by a connection to `wikileaks.org`, a USB storage unmount event, and a user session logoff.
- **Analytic Finding:** Primary and secondary threat conditions coincide with a cohesive, chronologically confirmed data exfiltration sequence.

- **Logged Decision:**
  - **Operational Disposition:** `Confirm Threat`
  - **Day Scenario:** `S-1 (IP Theft via Media)`

6. Case Assessment & Submission Protocol

After evaluating each flagged date individually, scroll to the **Case Assessment** module at the base of the interface:

1. **Completeness Verification:** Verify that every flagged date in the *Detected Anomalies* table has a logged operational disposition. *(For this training module, remaining dates may be marked Dismiss Alarm / Pass as non-binding demonstration placeholders).*
2. **Overall Case Closure Classification:** Assign the definitive case disposition for User 162 (**S-1: IP Theft via Media**, driven by the confirmed October 26 sequence).
3. **Analyst Case Justification Narrative:** Record a concise synthesis identifying the confirmed primary indicators, corroborated secondary deviations, presence/absence of storage media, and temporal timeline evidence.
4. **Analyst Metadata:** Enter your assigned **SOC Analyst ID** and assign your **Analyst Confidence Level** (1–5, or *Pass*).
5. **Subjective Reflections:** Complete the brief workload scales (Mental Demand and Frustration Level) and the interface evaluation fields.
6. **Evidence Export:** Click **"Export Forensic Evidence Package"**. The system will compile all timing data, row dispositions, and narrative justifications into the research database.